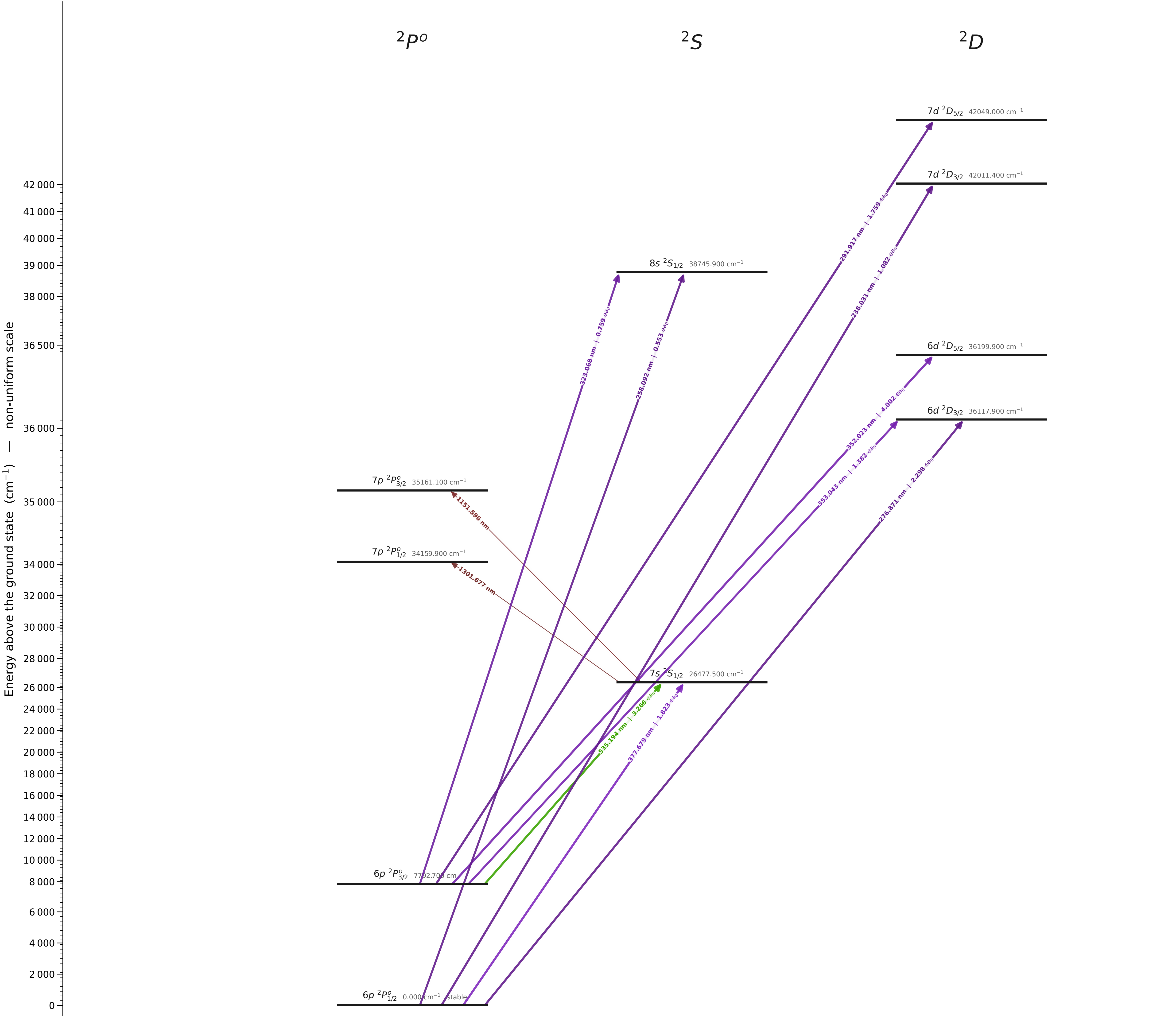


Thallium

Tl (Tl I). Strongest classified transitions below 2 μm (NIST): 10 levels, 11 lines.

Tl⁺ ionisation limit 49266.66 cm^{−1} = 6.10829 eV (202.977 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6p ² P _{1/2} ^o – 7d ² D _{3/2}	238.0306	237.9580	1259.4701	1.082	4.400e+07	-	-	-
6p ² P _{1/2} ^o – 8s ² S _{1/2}	258.0918	258.0146	1161.5729	0.553	1.800e+07	-	-	-
6p ² P _{1/2} ^o – 6d ² D _{3/2}	276.8710	276.7893	1082.7874	2.298	1.260e+08	-	-	-
6p ² P _{3/2} ^o – 7d ² D _{5/2}	291.9171	291.8317	1026.9780	1.759	4.200e+07	-	-	-
6p ² P _{3/2} ^o – 8s ² S _{1/2}	323.0684	322.9751	927.9536	0.759	1.730e+07	-	-	-
6p ² P _{3/2} ^o – 6d ² D _{5/2}	352.0234	351.9228	851.6264	4.002	1.240e+08	-	-	-
6p ² P _{3/2} ^o – 6d ² D _{3/2}	353.0425	352.9416	849.1681	1.382	2.200e+07	-	-	-
6p ² P _{1/2} ^o – 7s ² S _{1/2}	377.6792	377.5719	793.7755	1.823	6.250e+07	-	-	-
6p ² P _{3/2} ^o – 7s ² S _{1/2}	535.1944	535.0456	560.1562	3.266	7.050e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation
Level caption: NIST energy and measured lifetime (where one exists).